Profiling Donors of Blood, Money, and Time

A SIMULTANEOUS COMPARISON OF THE GERMAN POPULATION

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Understanding donor profiles is crucial for donor relationship management. Whereas previous research has focused on profiling blood, money, or time donor segments separately, we define seven donor profiles based on their former donation behavior for blood, money, and time donation and compare them to non-donors. Relying on representative data from the German Socio Economic Panel, we use an extensive set of characteristics that include sociodemographic, psychographic, health-related, and geographical measures and simultaneously investigate profiles of donors for single and multiple donation forms and non-donors by means of a multinomial logistic model. Our results reveal valuable insights for donor acquisition and retention strategies of nonprofit organizations along the identified profiling characteristics of donor segments. By this, our findings help nonprofit organization managers to better target single and multiple donors across three donation forms.

Keywords: *donor segments, donor relationship management, donor profiles, donor typologies*

NONPROFIT ORGANIZATIONS (NPOS) THAT SEEK VOLUNTEERS, money, or blood donors are facing fiercer competition (Naskrent and Siebelt 2011). In the United States, for example, the number of NPOs increased from 12,000 in 1940 to approximately 1.23 million in 2005, although the number of donors has been increasing underproportionally to the number of NPOs (Venable and others 2005).

As a response to the increasingly competitive market environment, NPOs are continually adopting marketing instruments (Eikenberry 2009; Sargeant, Hudson, and West 2008; Sosin 2012). Specifically, NPOs are aggressively relying on direct mail marketing strategies leading to the NPO sector to be among the top five industries employing direct mailings (Feld and others 2013). For example, in November 2006, more than fifteen German NPOs sent more than one million direct mail articles to (potential) donors (Hermes 2007). Because of the competitive donor relationship activities employed by NPOs and to natural restrictions of donors' time, money, and blood budgets, NPOs engaged in different forms of giving are increasingly competing with each other for gaining donors' attention and loyalty. Indeed,

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studies have shown that donors' main reason for lapsing after their first donations is switching to other donation forms (Sargeant 2001).

Against this background, a thorough understanding of (potential) donor profiles across different donation forms is essential because of its importance for successful targeting and relationship donor management (Naskrent and Siebelt 2011; Einolf 2011; Meslin, Rooney, and Wolf 2008), especially because former donor engagement for different donation forms is related to donor loyalty (Sargeant 2001). Although donor profiling for single donation forms that is, blood, money, or volunteering, has been addressed by several studies in the past, only single studies (for example, Houston 2006) analyze drivers of different donation types. Still, even these single existing studies that investigate different forms of donation analyze each donation form in a separate model. Consequently, these studies are not able to differentiate between segment profiles for donors who are engaged only into one form of donation (for example, blood doors) and those engaged into more than one donation form (for example, blood and money donors). A differentiated analysis of donor segments based on their engagement for one or multiple donation forms becomes even more important in light of recent empirical studies evidencing that a remarkable portion of donors engages in more than one form of donation (Bekkers 2006; Priller and Schupp 2011).

From a research perspective, differentiating between single-donation and multiple-donation donor segments based on past donation behavior and investigating their profiles simultaneously provides additional insights that cannot be revealed in case each donation form is investigated separately. Profiles of individuals engaged in multiple donation forms are likely to differ from those of the broader single-donor segment. On the one hand, multiple donation engagement is positively related to higher donor commitment (Sargeant and Woodliffe 2005), which is essential for donor retention and loyalty (Sargeant 2001). On the other hand, multiple donors face time and resource limitations. Consequently, it is important to understand how profiles of these "multiple donors" differ from those of single-form donors or non-donors.

From a managerial perspective, this question is also substantial for efficient donor targeting: Both donor retention and new donor acquisition require that segment profiles are described as accurately as possible. For example, from the perspective of blood donation services interested in new donor acquisition, grouping all individuals who do not donate blood into a non-donor group is insufficient because subsegments may exist that engaged formerly or still engage in other donation forms such as monetary donation, voluntary work (time donation), or both. These individuals may have higher adeptness to donation in general and consequently to blood donation. Indeed, past donation behavior is a strong predictor of future donation behavior because prosocial behavior becomes a habit that drives the likelihood of re-donation (Rosen and Sims 2011). In addition, prior behavior has a self-commitment impact that increases the probability of a future donation, even in other donation forms (Monin and Miller 2001; Mount 1996). These arguments suggest that specific donor relationship programs, depending on donors' engagement for different donation forms, can be advantageous for targeting strategies of NPOs. Especially NPOs engaged in multiple donation forms (for example, the Red Cross, which collects blood and money and seeks volunteers) could strongly benefit from differentiated targeting strategies for donor subsegments based on their former behavior for donation forms.

Our article addresses profiling of (former) donors engaged in single and multiple donation forms and contributes to the current scientific discourse as follows: (1) We examine the profiles of (former) blood, money, and time donors simultaneously and identify profiles of multiple- versus single-donor segments as compared to non-donors. By this, we add to existing research by defining donor segments based on their former donation behavior for different donation forms and profiling these segments simultaneously. We apply a multinomial logistic model and investigate whether individuals who formerly or actively engage in more than one donation form differ from non-donors or single donation form donors. Grounding on former donation behavior we define seven segments (blood, money, time donors, as well as their combinations) and identify segment-specific profiles compared to the non-donor segment. (2) We use a wide set of predictors including sociodemographic, psychographic, health-related as well as geographical variables for profiling donors. Thus, our results provide specific insights regarding donor profiling. (3) We use data from the German sociodemographic panel made up of more than 11,000 individuals, which is representative for the German population. To the best of our knowledge, our study is the first to compare donor segments based on the former donation behavior for blood, monetary, and time donations for the German market. We can extend findings from other countries, for example, the Netherlands (Bekkers 2006) and Canada (Rajan, Pink, and Dow 2009) by our findings from the German donation population.

Additionally, our research has substantial managerial relevance for NPOs operating within blood, money, and time donation sectors, especially for those NPOs that operate in more than one donation sector. We offer distinguished implications with respect to targeting options for new and existing donors depending on their engagement in other donation forms.

Literature Review

In the following we give an overview of existing studies investigating drivers of blood, money, and time and discuss their findings.

Overview of Studies on Blood, Money, and Time Donations

We identify a large body of research on the determinants of blood donation, money donation, and volunteer behavior. We analyze thirty-eight studies on the drivers of donation behavior with respect to blood, money, and time according to two dimensions: (1) donation forms considered and (2) predictor categories included in the analyses (see Table 1 for an overview).

With respect to the first dimension, our literature review shows that former studies focus mostly on a single donation form, that is, money, blood, or volunteering (time donation). Few studies consider two donation forms, for example, money and blood donation (Bekkers 2006; Priller and Schupp 2011) or money and time donation (Bekkers 2010; Berger 2006; Einolf 2011; Mesch et al. 2006), and we can identify only two studies that analyze drivers of all three donation forms—blood, money, and volunteering (time; Houston 2006; Lee, Piliavin, and Call 1999).

Independently of the number of considered donation forms, existing studies analyze drivers for each donation form separately, mainly by using bivariate logit regression analyses for the identification of drivers of donation behavior (Taniguchi 2010; Wiepking 2007; Wiepking and Bekkers 2010) and defining non-overlapping donor segments (Houston 2006). This underlines the need for a simultaneous consideration of engagement for different donation

Table 1. Literature Review of Studies on Blood, Money, and Time Donation

				Dependent Variables			*Factors*				
Variable Type		*Empirical Study*	*Data*	*Blood*	*Money*	*Time*	*Behavioral*	*Sociodemo-graphic*	*Psychographic*	*Health*	*Geographical*
Blood donors vs. non-donors	1	Duboz et al. (2010)	*N*=600	x				x			
	2	Solomon (2012)	*N*=195	x				x			
	3	Burnett (1981)	*N*=577	x				x	x		
	4	Nonis et al. (1996)	*N*=195	x				x	x		
	5	Tscheulin and Lindenmeier (2005)	*N*=210	x				x	x		
	6	Andaleeb and Basu (1995)	*N*=148	x				x	x	x	
	7	Boulware et al. (2002)	*N*=385	x				x	x	x	
	8	Newman and Pyne (1997)	*N*=236	x				x	x	x	
Blood donor types	9	Schreiber et al. (2005)	*N*=179,409	x				x			
	10	Duboz and Cunéo (2009)	*N*=1,400	x				x		x	
	11	Schreiber et al. (2003)	*N*=6.4 million	x				x		x	
	12	Ibrahim and Mobley (1993)	*N*=521	x				x	x	x	
	13	Burnett and Leigh (1986)	*N*=945	x				x	x		
	14	Ownby et al. (1999)	*N*=879,816	x				x			x
	15	Thomson et al. (1998)	*N*=34,726	x			x	x			x
	16	Veldhuizen et al. (2009)	*N*=340,470	x			x	x			x
Money donors vs. non-donors & donation amount	17	Banks and Tanner (1999)	*N*=85,000		x			x			
	18	Hughes and Luksetich (2008)	*N*=2,041		x			x			
	19	Piper and Schnepf (2008)	*N*=12,679		x			x			
	20	Kottasz (2004)	*N*=217		x			x	x		
	21	Wiepking and Bekkers (2010)	*N*=1,101		x			x	x		
	22	Schervish and Havens (1997)	*N*=2,065		x		x	x	x		
Money donation types	23	Wiepking (2007)	*N*=1,316		x		x	x	x		
	24	Rajan, Pink, and Dow (2009)	*N*=13,125		x		x	x	x	x	x
Time donors vs. non-donors & donation amount	25	Jeong (2010)	*N*=1,008			x		x	x	x	
	26	Taniguchi (2010)	*N*=2,836			x		x	x		x
	27	Reed (2000)	*N*=18,301			x	x	x	x		
Donation type	28	Taniguchi and Thomas (2011)	*N*=1,612			x		x	x		
Blood & money donors	29	Priller and Schupp (2011)	*N*=16,963	x	x		x	x	x		
	30	Bekkers (2006)	*N*=1,140	x	x			x	x	x	x
Money & time donors	31	Mesch et al. (2006)	*N*=532		x	x		x			
	32	Berger (2006)	*N*=14,724		x	x			x		
	33	Einolf (2011)	*N*=3,032		x	x		x	x		
	34	Bryant (2003)	*N*=1,481		x	x		x	x		x
	35	Jones (2006)	*N*=2,719		x	x	x	x	x		
	36	Bekkers (2010)	*N*=1587		x	x	x	x	x	x	x
Blood, money, & time donors	37	Houston (2006)	*N*=1,301 & *N*=1,142	x	x	x		x			
	38	Lee et al. (1999)	*N*=1,002	x	x	x	x		x		
This study			*N*=12,487	x	x	x	x	x	x	x	x

forms based on former donation behavior. By this, profiling donor segments engaged in more than one donation form can be defined and compared to single-donation-form donors or non-donors. Donors are confronted with trade-offs regarding their time, money and blood. Consequently, their engagement with respect to one specific donation form is likely to depend upon donation behavior regarding the other forms. These interrelationships, which are substantial for unbiased identification of donor profiles, are not sufficiently taken into account in previous studies. A more detailed analysis should therefore help to target these donor segments adequately.

Regarding the second dimension, previous studies use independent factors in addition to the donation forms already considered that may drive donor behavior. These considered independent factors can be classified into the following five categories: factors related to donation behavior (for example, Priller and Schupp 2011; Schervish and Havens 1997); sociodemographic factors (for example, Houston 2006; Wiepking 2007); psychographic factors (for example, reasons to donate, Burnett and Leigh 1986; Nonis et al. 1996; or personality traits, Bekkers 2006, 2010); health-related factors (for example, health status, Duboz and Cunéo 2009; Jeong 2010; or the presence of diseases, Boulware et al. 2002; Schreiber et al. 2003); and, last, geographical factors (for example, region, urbanization level). From our literature analysis of thirty-eight studies we identify that the majority of the studies (94.74 percent) include sociodemographic variables in their models, followed by psychographic variables (65.79 percent), health-related variables (26.32 percent), and variables related to donation behavior (26.32 percent). Geographical variables are underrepresented (used in 21.05 percent of our studies). Only three studies include factors from all five categories (Bekkers 2006, 2010; Rajan et al. 2009), indicating a need for additional research insights on the simultaneous influence of factors along all categories.

In summary, our literature review indicates the necessity for additional research along two dimensions: simultaneous consideration of (1) former donation behavior for all three donation forms—money, blood, and time—and (2) different predictor categories. We substantially expand existing findings by addressing both gaps: we consider typologies of donors based on their former donation behavior of money, blood, and time simultaneously, and we include factors from all five categories identified from the literature review in Table 1.

Drivers for Monetary, Blood, and Time Donations

Earlier research on determinants of blood, money, and time donations provides insights on drivers of each donation form. Tables 2 to 4 summarize the findings of our literature overview on the drivers for money, blood, and time donation. Within each table, studies are grouped according to whether they consider the binary donation decision or the donation amount as dependent variable. The columns display single drivers and the single-cell reference to a study that has evidenced a driver's effect (positive significant, negative significant, or not significant).

With respect to blood donation, the findings reveal that blood donors are typically male, educated, healthy, white, married, have children, and have a rare blood type (Andaleeb and Basu 1995; Bekkers 2006; Boulware et al. 2002; Duboz et al. 2010; Newman and Pyne 1997; Nonis et al. 1996; Priller and Schupp 2011). Moreover, the frequency of blood donation is significantly higher for male, older, white, and educated individuals (Burnett and Leigh 1986; Schreiber et al. 2005; Veldhuizen et al. 2009).

Table 2. Blood Donation: Causal Relationships

			Factors																	
			Behavioral						Sociodemographic											
			Blood donor	Blood donation frequency	Money donor	Money amount donated	Volunteer	Organ donor	Gender (male)	Age	Age²	Educational level	Employment	Income	Home ownership	Marital status (married)	Household size	Children	Religiousness	Church attendance
dependent variables: Blood	Blood donation	+			29				1,3, 4,6, 7,8	1,6, 30,37		1,3, 4,5, 29,30	29	7		3		3	3	
		–								2,5, 8,29	37	37		30						
		ns							5,8, 29,30,37	4,7		6,7	7, 30	6, 37	30	4,6, 7,37	6	37	7, 30	30, 37
	Blood donation frequency	+	14	16					12,13, 14,16	9,11, 14,16		9,11, 12,14		16	16	12		12		
		–												13		13				
		ns							9,11	12			12	12						

1. Duboz et al. (2010)
2. Solomon (2012)
3. Burnett (1981)
4. Nonis et al. (1996)
5. Tscheulin and Lindenmeier (2005)
6. Andaleeb and Basu (1995)
7. Boulware et al. (2002)
8. Newman and Pyne (1997)
9. Schreiber et al. (2005)
10. Duboz and Cunéo (2009)
11. Schreiber et al. (2003)
12. Ibrahim and Mobley (1993)
13. Burnett and Leigh (1986)
14. Ownby et al. (1999)
15. Thomson et al. (1998)
16. Veldhuizen et al. (2009)
17. Banks and Tanner (1999)
18. Hughes and Luksetich (2008)
19. Piper and Schnepf (2008)
20. Kottasz (2004)
21. Wiepking and Bekkers (2010)
22. Schervish and Havens (1997)
23. Wiepking (2007)
24. Rajan et al. (2009)
25. Jeong (2010)
26. Taniguchi (2010)
27. Reed (2000)

Money donors tend to be female, older, employed, and home owners with high incomes and academic degrees (Banks and Tanner 1999; Bekkers 2010; Einolf 2011; Houston 2006; Jones 2006; Piper and Schnepf 2008; Wiepking 2007). Furthermore, the amount of donated money increases with age, educational level, income, and religious behavior (Banks and Tanner 1999; Mesch et al. 2006; Rajan et al. 2009; Wiepking and Bekkers 2010).

Volunteer profiles are similar to the profiles of money donors with respect to gender (mostly female), education (higher degree of education, and higher income) (Einolf 2011; Houston 2006; Jeong 2010; Taniguchi and Thomas 2011). Similarly, money donors and volunteers are often religious and have children (Bekkers 2010; Houston 2006; Jeong 2010; Jones 2006;

Table 2. Blood Donation: Causal Relationships (*Continued*)

Factors																									
Socio-demographic		*Psychographic*														*Health*						*Geographical*			
Race (white)	*Nationality (German)*	*Risk taking*	*Health salience*	*Satisfaction*	*Happiness*	*Reciprocity*	*Reward/Incentive*	*Altruism*	*Subjective norm*	*Self esteem*	*Requests for donations*	*Helpfulness*	*Empathy*	*fear*	*political interest*	*blood type (rare)*	*rh type*	*state of health*	*prior hospital contact*	*health insurance (private)*	*health risk*	*country of birth (North America)*	*region*	*residence change*	*urbanization level*
7,8		6	3		29	29	4,5,8	8	8			30	30			3,4		8,30		7					
	29	3								3				5,7,8							6		29		
37				29		29				6								7	7		4				30
9,11,14		13		12				13								13	14	11				14			16
16			13							13				12											
																			12						

28. Taniguchi and Thomas (2011)
29. Priller and Schupp (2011)
30. Bekkers (2006)
31. Mesch et al. (2006)
32. Berger (2006)
33. Einolf (2011)
34. Bryant (2003)
35. Jones (2006)
36. Bekkers (2010)
37. Houston (2006)
38. Lee et al. (1999)

+: positive effect

–: negative effect

ns: not significant

Mesch et al. 2006; Reed 2000; Taniguchi and Thomas 2011). However, with respect to the number of volunteered hours, there are no distinct tendencies except that highly educated people spend more time volunteering (Mesch et al. 2006; Taniguchi 2010).

The majority of discovered effects regarding the drivers of blood, money, and time donation are often ambiguous across studies (see Tables 2–4). For example, the effect of age on blood donation is not determined consistently because some studies reveal blood donors to be older (Andaleeb and Basu, 1995; Bekkers 2006; Duboz et al. 2010), whereas other studies demonstrate that blood donors are younger (Newman and Pyne 1997; Priller and Schupp 2011; Solomon 2012). Boulware et al. (2002) find that blood donors have higher incomes, whereas Bekkers (2006) reaches the

Table 3. Money Donation: Causal Relationships

			Factors																	
			Behavioral						*Sociodemographic*											
			Blood donor	*Blood donation frequency*	*Money donor*	*Money amount donated*	*Volunteer*	*Organ donor*	*Gender (male)*	*Age*	*Age²*	*Educational level*	*Employment*	*Income*	*Home ownership*	*Marital status (married)*	*Household size*	*Children*	*Religiousness*	*Church attendance*
Dependent Variables: Money	money donation	+	29		36	36	23		24,29	17,23,29, 31,33, 35,37		17,29, 30,31 33,35, 36,37	17,33	17,30, 31,33,37	17,23			17	33,36	23
		–					24,36		19,35, 36,37	24,36	19	24		24		24			24	
		ns	36					36	23,30	23,30,33		23	29,30, 33	23	23,30, 36	19,37	17,23, 24	33,35, 37	30,33, 36	23,30, 36,37
	money amount donated	+			22		22		30	17,21, 22,23, 30,31		17,21, 22,23, 24,30, 31,35	17	17,18, 30,31	17,21	19,21		21	21,22, 30	21,22, 23,30, 35
		–					24		19	24				23		24	17, 22			
		ns					23,35		21,23, 24			23	30	21,22 24	23,30		23,24	17	22,24	

1. Duboz et al. (2010)
2. Solomon (2012)
3. Burnett (1981)
4. Nonis et al. (1996)
5. Tscheulin and Lindenmeier (2005)
6. Andaleeb and Basu (1995)
7. Boulware et al. (2002)
8. Newman and Pyne (1997)
9. Schreiber et al. (2005)
10. Duboz and Cunéo (2009)
11. Schreiber et al. (2003)
12. Ibrahim and Mobley (1993)
13. Burnett and Leigh (1986)
14. Ownby et al. (1999)
15. Thomson et al. (1998)
16. Veldhuizen et al. (2009)
17. Banks and Tanner (1999)
18. Hughes and Luksetich (2008)
19. Piper and Schnepf (2008)
20. Kottasz (2004)
21. Wiepking and Bekkers (2010)
22. Schervish and Havens (1997)
23. Wiepking (2007)
24. Rajan, Pink and Dow (2009)
25. Jeong (2010)
26. Taniguchi (2010)
27. Reed (2000)

opposite conclusion. Similar ambiguous results are also revealed with respect to other independent variables, for example, political interest (Jeong 2010; Rajan et al. 2009), household size (Andaleeb and Basu, 1995; Banks and Tanner, 1999; Rajan et al. 2009; Schervish and Havens, 1997; Wiepking 2007), blood type (Burnett and Leigh, 1986), as well as mood and mindset (Andaleeb and Basu 1995; Burnett 1981; Burnett and Leigh 1986; Priller and Schupp 2011).

In summary, previous research has addressed drivers of blood, money, and time donation. Still, some results are not consistent across studies (see Tables 2–4). Our data set and methodological approach allow for simultaneously analyzing drivers of donor segments that are

Table 3. Money Donation: Causal Relationships (*Continued*)

Factors																									
Sociodemographic		*Psychographic*														*Health*						*Geographical*			
Race (white)	*Nationality (german)*	*Risk taking*	*Health salience*	*Satisfaction*	*Happiness*	*Reciprocity*	*Reward/incentive*	*Altruism*	*subjective norm*	*Self-esteem*	*Requests for donations*	*Helpfulness*	*Empathy*	*Fear*	*Political interest*	*Blood type (rare)*	*rh type*	*State of health*	*Prior hospital contact*	*Health insurance (private)*	*Health risk*	*Country of birth (North America)*	*Region*	*Residence change*	*urbanization level*
35				29	29	29					23		30,36										29		
	29			24		29						36			24										30
19,31												30						24,30, 36		23		24			36
											23	35	30					30		23					
																									30
22,31				24							23	30			24			24		23		24			

28. Taniguchi and Thomas (2011)
29. Priller and Schupp (2011)
30. Bekkers (2006)
31. Mesch et al. (2006)
32. Berger (2006)
33. Einolf (2011)
34. Bryant (2003)
35. Jones (2006)
36. Bekkers (2010)
37. Houston (2006)
38. Lee et al. (1999)

+: positive effect

–: negative effect

ns: not significant

engaged in different forms of donation, that is, blood, money, time, and combinations of these forms. We use a wide range of predictors from all previously described categories and identify the characteristics of donor segments that are involved in all three forms and other segments that choose a single favored donation format.

Sample

The German Socio-Economic Panel Study (SOEP)[1] is a representative longitudinal study of German private households and individuals and represents one of the most important

Table 4. Time Donation: Causal Relationships

Dependent Variables			*Factors: Behavioral*						*Factors: Sociodemographic*											
			Blood donor	*Blood donation frequency*	*Money donor*	*Money amount donated*	*Volunteer*	*Organ donor*	*Gender (male)*	*Age*	*Age²*	*Educational level*	*Employment*	*Income*	*Home ownership*	*Marital status (married)*	*Household size*	*Children*	*Religiousness*	*Church attendance*
Time	Volunteering	+	36		36		27,36		25,33	26,35		25,27, 28,31, 33,35, 36,37		31,37		28		27,28, 33,35, 37	25,36, 37	28,37
		–							28,35, 36,37	33,36	28		33		36					
		ns				36		36	28	25,31, 37			25	25,33, 35		25,37		25,28	25,33, 36	36
	Volunteering hours	+				36		36		26		26,31	26			26			26	
		–					35				26									35
		ns							26,33	31		35		26,31, 35				26		

1. Duboz et al. (2010)
2. Solomon (2012)
3. Burnett (1981)
4. Nonis et al. (1996)
5. Tscheulin and Lindenmeier (2005)
6. Andaleeb and Basu (1995)
7. Boulware et al. (2002)
8. Newman and Pyne (1997)
9. Schreiber et al. (2005)
10. Duboz and Cunéo (2009)
11. Schreiber et al. (2003)
12. Ibrahim and Mobley (1993)
13. Burnett and Leigh (1986)
14. Ownby et al. (1999)
15. Thomson et al. (1998)
16. Veldhuizen et al. (2009)
17. Banks and Tanner (1999)
18. Hughes and Luksetich (2008)
19. Piper and Schnepf (2008)
20. Kottasz (2004)
21. Wiepking and Bekkers (2010)
22. Schervish and Havens (1997)
23. Wiepking (2007)
24. Rajan et al. (2009)
25. Jeong (2010)
26. Taniguchi (2010)
27. Reed (2000)

tools for social and economic research in Germany (Schupp and Wagner 2002). The panel study has been conducted annually since 1984 at the German Institute for Economic Research, DIW Berlin. Every year, 11,000 households and more than 20,000 individuals are invited by the market research institute TNS "*Infratest Sozialforschung*" [Infraset] and participate in the paper-and-pencil survey (Krupp 2007). The German Socio Economic Panel (SOEP) has a high stability over time: 53 percent of households that participated in the first SOEP wave were still participating twenty-five years later in the SOEP wave of 2010.

The survey contains information on different topics including health and satisfaction indicators, household composition, occupational biographies, and different psychographic aspects. Because the survey is continually being adapted to current social developments, certain questions on specific topics are only included at irregular intervals ranging from two to ten years.

Table 4. Time Donation: Causal Relationships (*Continued*)

Factors																									
Sociodemographic		*Psychographic*														*Health*						*Geographical*			
Race (white)	*Nationality (German)*	*Risk-taking*	*Health salience*	*Satisfaction*	*Happiness*	*Reciprocity*	*Reward/incentive*	*Altruism*	*Subjective norm*	*Self esteem*	*requests for donations*	*Helpfulness*	*Empathy*	*Fear*	*Political interest*	*Blood type (rare)*	*rh type*	*State of health*	*Prior hospital contact*	*Health insurance (private)*	*Health risk*	*Country of birth (North America)*	*Region*	*Residence change*	*Urbanization level*
31,35												36	36												
37				25											25			25,36							36
																									26
31												35													

28. Taniguchi and Thomas (2011)
29. Priller and Schupp (2011)
30. Bekkers (2006)
31. Mesch et al. (2006)
32. Berger (2006)
33. Einolf (2011)
34. Bryant (2003)
35. Jones (2006)
36. Bekkers (2010)
37. Houston (2006)
38. Lee et al. (1999)

+: positive effect

-: negative effect

ns: not significant

Questions on donation behavior of individuals belong to these specific topics and are not part of every survey wave. Specifically, questions on blood and monetary donation behavior were integrated for the first time into the 2010 survey wave (see Table 5 for the detailed questions) and are part of every fifth survey wave. Questions on time donation (volunteering) were part of the 2009 survey wave and are included every two years in the survey. This specific structure combines both survey waves 2009 and 2010 in order to consider former donation behavior for all three donation forms, that is, blood, money, and time. Consequently, participants of both SOEP waves 2009 and 2010 serve as the initial basis for our empirical analyses.

The resulting sample comprises 18,024 participants (18,587 respondents participated in the 2009 wave and 19,101 respondents participated in the 2010 wave). To reduce biases because of external factors and because we aim to analyze only drivers of potentially eligible donors

Table 5. Operationalization of Variables

	Variable		Operationalization
Dependent / Behavioral	Blood donation	Have you donated blood in the last 10 years?	Dummy variables 1 = Yes; 0 = No
	Money donation	And now a question about your donations. We understand donations here as giving money for social, church, cultural, community, and charitable aims, without receiving any direct compensation in return. These donations can be large sums of money but also smaller sums, for example, the change one puts into a collection box. We also count church offerings. Did you donate money last year, in 2009—not counting membership fees?	
	Time donation	Which of the following activities do you take part in during your free time? Volunteer work in clubs or social services	
	Donor segments	Defined based on questions on blood, money and time donations	1 = Only blood donation 2 = Only money donation 3 = Only time donation 4 = Blood and money donation 5 = Money and time donation 6 = Blood and time donation 7 = blood, money, and time donation
Sociodemographic	Gender (Male)	Gender	Dummy variable 1 = Male; 0 = Female
	Age	Age	18 to 71
	Education/ Apprenticeship	Are you currently in some sort of education? In other words, do you attend a school or institution of higher education (including doctorate/PhD), are you engaged in an apprenticeship, or are you participating in further education or training?	Dummy variable 1 = Yes; 0 = No
	Full-time employment	Are you currently in paid employment? Which of the following applies best to your status? . . . Full-time employed	Dummy variable 1 = Yes; 0 = No
	Unemployment	Are you officially registered as unemployed at the Employment Office ("*Arbeitsamt*")?	Dummy variable 1 = Yes; 0 = No
	Net income	How high was your income from employment last month? . . . Net income, which means the sum after deduction of taxes, social security, and unemployment and health insurance	
	Marital status (Married)	What is your marital status? . . . Married	Dummy variable 1 = Yes; 0 = No
	Children	Do children who were born in 1994 or later live in your household?	Dummy variable 1 = Yes; 0 = No
	Inheritance	Did you or another member of the household receive a large sum of money or other forms of wealth (car, house, and so on) as inheritance, gift, or lottery winnings last year? We refer to money or other forms of wealth worth more than 500 euros. As: Inheritance	Dummy variable 1 = Yes; 0 = No

Table 5. Operationalization of Variables (*Continued*)

	Variable		*Operationalization*
Sociodemographic	Gift	Did you or another member of the household receive a large sum of money or other forms of wealth (car, house, and so on) as inheritance, gift, or lottery winnings last year? We refer to money or other forms of wealth worth more than 500 euros. As: gift	Dummy variable 1 = yes; 0 = no
	Lottery winnings	Did you or another member of the household receive a large sum of money or other forms of wealth (car, house, and so on) as inheritance, gift, or lottery winnings last year? We refer to money or other forms of wealth worth more than 500 euros. As: lottery winnings	Dummy variable 1 = yes; 0 = no
Psychographic	Risk taking	How do you see yourself: Are you generally a person who is fully prepared to take risks or do you try to avoid taking risks?	0 = Risk averse to 10 = Fully prepared to take risks
	Job satisfaction	How satisfied are you today with the following areas of your life? How satisfied are you with . . . . . .your job/your income/your health	0 = Totally unhappy to 10 = Totally happy
	Free-time satisfaction	How satisfied are you today with the following areas of your life? How satisfied are you with . . . . . .your free time/your family life.	0 = Totally unhappy to 10 = Totally happy
	Life satisfaction	In conclusion we would like to ask you about your satisfaction with your life in general: How satisfied are you with your life, all things considered?	0 = Completely dissatisfied to 10 = Completely satisfied
	Mood	I will now read to you a number of feelings. Please indicate for each feeling how often or rarely did you experience this feeling in the last four weeks. How often have you felt . . . . . .angry/worried/happy/sad?	1 = very rarely 2 = rarely 3 = occasionally 4 = often 5 = very often
	Interest in politics	Generally speaking, how interested are you in politics?	1 = not at all interested 2 = not so interested 3 = interested 4 = very interested
Health	Care-dependent individuals in the household	Does someone in your household need care or assistance on a constant basis because of age,sickness, or medical treatment?	Dummy variable 1 = yes; 0 = no
	Chronic illness	Are you suffering for at least one year, or as a chronic condition, from certain complaints or illnesses?	Dummy variable 1 = yes; 0 = no
	Health insurance (compulsory)	How are you ensured for sickness: Do you have state health insurance or are you almost exclusively privately ensured? . . .In compulsory health insurance	Dummy variable 1 = yes; 0 = no
Geographical Health Data	New states of Germany	Former East Germany	Dummy variable 1 = yes; 0 = no

we exclude individuals who are not permitted to donate blood for German legal reasons. First, we exclude 3,184 individuals who are younger than eighteen, or who are seventy-one or older. Second, we exclude 3,498 persons who are not permitted to donate blood because of self-reported medical reasons. Our final data set consists of 12,487 (potential) donors.

Data

This section provides the measures and descriptive statistics for the dependent and independent variables of our model. Table 5 provides a detailed overview of the original SOEP questions and operationalization of the variables (see Appendix A for an overview of descriptive statistics). Past donation behavior is used to define donor segments and to create our dependent variable. The dependent variable consists of eight categories that indicate one of the eight groups outlined previously, with non-donors serving as the reference category. We include explanatory variables from the remaining four classes described earlier: sociodemographic, psychographic, health-related, and geographical factors.

Donation Behavior

We include three indicators of (former) donation behavior for blood, money, and time. In the survey wave of 2010, respondents were asked directly whether they had donated blood in the ten years prior to the study or specifically in 2009. Consequently, we define donors as individuals who have donated during the period 1999 to 2009.[2] Similarly, donors are asked in the SOEP questionnaire 2010 whether they have donated money in 2009. We define former monetary donation based on this question. Specifically, respondents answering the question with yes are considered to be former monetary donors. Time donors (volunteers) are defined according to the survey wave of SOEP 2009, in which respondents were asked whether they take part in volunteer work at clubs or with social services in the current year (that is, 2009). Based on these three questions on former donation behavior out of two survey waves of 2009 and 2010, we define seven donor segments engaged in single or multiple donation forms as well as non-donors. However, although the SOEP data allows combining information on former donation behavior, information on former blood, time, and monetary donation behavior refers to different time periods. Specifically, the monetary and time donation behavior refers to 2009, while the blood donation behavior refers to the period 1999–2009. This different time scaling of former behavior for blood, time, and money donation implies restrictions on chronological order of donation engagement for different donation forms. Still, the segment definition based on former donation behavior corresponds to managerial practice as donor relationship managers usually use all information on former donorship that is available in their databases and enrich this information with external data. Data from different internal or external sources contain historical information with different time scaling, for example, a NPO operating in the blood donation sector may have full historical information on blood donation behavior, while information on volunteering engagement may be based on a cross-section marketing research survey and be available only for the point of time when the survey was conducted. This logic supports our segment definition despite the restriction of different time scalings for former donation of money and time on one hand and blood on the other. Besides, the segment definition of blood donors on the ten years prior to the study addresses the generally higher re-donation probability of former donors (Rosen and Sims 2011).

We identify eight donor segments that were formerly engaged in (1) blood donation, (2) money donation, (3) time donation, (4) blood and money donation, (5) money and time donation, (6) blood and time donation, (7) all three donation forms, and (8) none (non-donors; see Figure 1). We derive a variable defining all groups, with non-donors as the reference category, which serves as our dependent variable in the multinomial logistic regression.

Sociodemographic Factors

Based on previous findings we include variables that were found to determine donation behavior significantly. Thus, we account for respondent gender (gender; Nonis et al. 1996; Tscheulin and Lindenmeier 2005) and family environment (marital status and children; Burnett 1981) in our model. Additionally, we control for respondent employment status (Bekkers 2006; Priller and Schupp 2011), full time, employed. or unemployed and use part-time employment as the base category. Last, we control for the professional education level of respondents (education/apprenticeship).

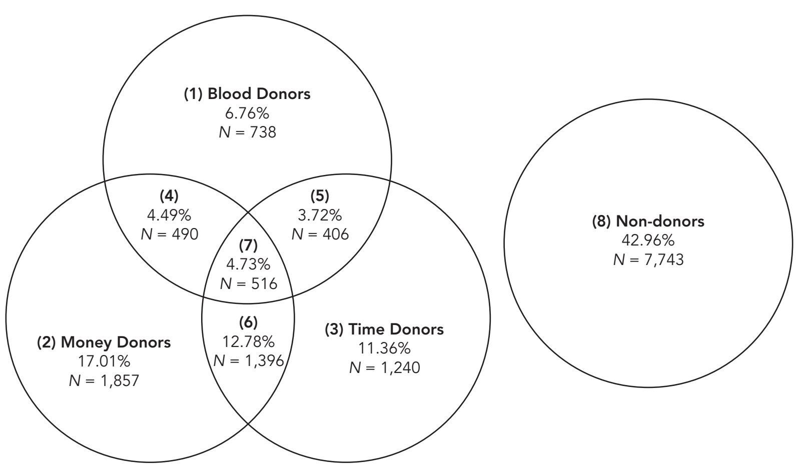
Our sample shows a balanced structure with respect to gender with 49.40 percent male participants. We detect that 47.91 percent of individuals are employed full time and that 11.04 percent were in training at the time of the survey. These results are consistent with the quota of unemployed respondents (7.08 percent). We also account for age specific effects (age) because previous studies identify significant age effects (for example, Andaleeb and Basu 1995; Newman and Pyne 1997).

Income was identified as a relevant predictor particularly for money donations (Banks and Tanner, 1999; Boulware et al. 2002; Jones 2006; Mesch et al. 2006; Rosen and Sims 2011). We include in our study the variables net income, inheritance, gift and lottery winnings and consider the effect of the general financial circumstances of respondents.

Psychographic Factors

Blood donation is related to psychological barriers, for example, fear of needles (Oswalt and Napoliello 1974). Therefore, consistent with past studies (Andaleeb and Basu 1995) we include risk-taking propensity in our study (risk-taking), as well as the general mood of the individual (mood), which is a factor made up of four emotional moods (angry, worried,

Figure 1. Overview of Donor Segments



happy, and sad). This factor is extracted by means of principal component analyses (α = 0.69, explained variance = 50 percent). A high value indicates a negative (sad) mood, and low values are an indicator of happiness or of a good mood prior to the survey.[3]

Individuals' satisfaction was shown to discriminate between donors and non-donors (Ibrahim and Mobley 1993; Nguyen et al. 2008; Priller and Schupp 2011; Rajan et al. 2009). Hence, we include a measure of overall life satisfaction (life satisfaction). In addition, we compute two indices for job satisfaction and leisure time satisfaction by means of a principal component analysis (α = 0.69, explained variance = 63.5 percent). The job satisfaction factor comprises the SOEP-variables with respect to satisfaction with work, income and health, and the free time satisfaction factor comprises satisfaction with family life and free time. Finally, we include the variable interest in politics, which measures the level of political engagement of respondents (Jeong 2010; Rajan et al. 2009).

Health-Related Factors

This group comprises characteristics related to health status as well as health insurance. We include the variable *"chronic illness"* as a predictor because previous studies have shown that blood donors generally have better health status (for example, Bekkers 2006; Newman and Pyne 1997). Additionally, the inclusion of variables related to health insurance is important (Boulware et al. 2002). Therefore, to capture systematic differences related to health insurance status, we include the variable compulsory health insurance, which indicates whether a person has a normal health insurance tariff. The majority of respondents in our sample (84.14 percent) have compulsory health insurance. Next, we consider whether respondent households include family members who need care (care-dependent individuals in the household), because these respondents (2.29 percent of the total sample) have a higher proneness toward helping others and show a higher commitment (Sargeant and Woodliffe 2005).

Geographical Factors

We consider geographical location effects between East and West Germany (new states of Germany; Priller and Schupp 2011).

Results

We estimate a multinomial logistic regression model with a nominal variable that displays our eight donor segments (including non-donors) as the dependent variable. Non-donors serve as the reference category. To facilitate a straightforward interpretation of the results, we display elasticity coefficients of independent variables that show the probability of change (in percentage) of a respondent belonging to one donor group, depending on a one percent change of an independent variable (Greene 2003).

The model is highly significant (Chi-square = 5,531, $p < .001$) with a pseudo-R^2 of .06 (Table 6). Other related studies that do not integrate individual motives and norms into their models report comparable levels of pseudo-R^2 statistics, especially for blood donation behavior (Houston 2006), indicating the validity of our model that captures the blood, money, and time donation behavior. Therefore, and because our objective is to identify segment profiles and not to predict segment assignments, the model is sufficiently appropriate for addressing our research question.

We identify different significant characteristics of all donor segments described previously.

Blood Donors

Age is shown to have the highest impact on blood donation behavior. Specifically, the probability of being a blood donor drops by 1.4 percent for every 1 percent increase in age (Table 6). Compulsory health insurance has the second largest elasticity, indicating that blood donors use compulsory health insurance. Moreover, donors are comparatively greater risk takers and less interested in politics. These findings add to findings of previous studies that show inconsistent findings on the effect of age and risk-taking (see Table 2). Finally, blood donors are typically not married and do not have children, which is consistent with the age effect but deviates from previous findings (see Table 2). These differences may be because of recent structural demographic changes in the German population. Furthermore, our blood donation segment comprises only blood donors who are not engaged in other donation forms, whereas past studies do not control for subsegments of multiple donors.

Money donors. The results show that the probability of monetary donations is significantly higher for female (potential) donors. In addition, money donors are shown to be significantly older with a higher net income as compared to non-donors, which is consistent to results of prior studies (see Table 3). Interestingly, we also discover that lottery winnings increase the probability to donate money. In contrast to blood donors, the probability of being involved only in money donation is higher when respondents have children. Regarding the health status, the likelihood of individuals donating money is significantly higher in case of chronically illnesses or if individuals do not rely on compulsory health insurance compared to non-donors. Money donors are more satisfied with their lives, have better short-term moods, and are comparatively interested in politics. These findings expand the existing research, because few studies have examined the effect of political interest, satisfaction, and happiness (see Table 3).

Time donors. Age is determined to be the variable with the highest impact with respect to donation of time, with an increasing probability for donation among younger people. Interestingly, contrary to prior findings (see Table 4), our results reveal time donors to be predominantly male, with lower incomes and a higher interest in politics. These differences may be due to the deviating segment definitions in our study as compared to former studies. Because of our distinct definition of donor segments, we are able to capture the effects of the independent variables with greater granularity as compared, for example, to studies that are based on bivariate segment definitions (Jeong 2010). Similarly to money donors, time donors do not rely on compulsory health insurance. Overall, people engaging in volunteer work are shown to have a higher level of satisfaction with their lives.

Blood and money donors. Consistent with blood donors, people who formerly engaged in blood and money donation are younger compared to non-donors. In contrast to the segment of blood donors, we find a higher probability of both (former) blood and money donation if the donor is female. This group has a higher income, is more interested in politics, and is also more often chronically ill as compared to non-donors. The latter may explain their involvement in both donation forms because individuals belonging to this segment may engage in money donation during times of illness as a substitute for blood donation.

Money and time donors. Individuals who formerly donated money and volunteered are shown to have a similar background. They are typically older, female, married, and have children. Furthermore, inheriting and receiving gifts increases the probability that an individual

Table 6. Elasticity Coefficients of Multinomial Logit Estimation

		Blood Donation			*Money Donation*			*Time Donation*		
	Variable	*Elasticity*	*Standard error*	*Significance*	*Elasticity*	*Standard error*	*Significance*	*Elasticity*	*Standard error*	*Significance*
Socio-demographic	Sex (male)	–0.013	0.041		–0.166	0.027	***	0.096	0.026	***
	Age	–1.418	0.171	***	0.917	0.114	***	–0.386	0.120	***
	Education/apprenticeship	–0.013	0.014		–0.017	0.014		0.025	0.009	***
	Full-time employment	0.041	0.053		–0.032	0.032		–0.004	0.038	
	Unemployement	0.013	0.010		–0.019	0.009	**	–0.031	0.009	***
	Net income	–0.020	0.051		0.169	0.025	***	–0.096	0.042	**
	Marital status (married)	–0.125	0.057	**	0.007	0.033		0.041	0.036	
	Children	–0.078	0.028	***	0.030	0.018	*	0.015	0.018	
	Inheritance	–0.008	0.006		0.001	0.003		–0.002	0.003	
	Gift	0.004	0.005		0.001	0.003		–0.010	0.004	**
	Lottery winnings	0.001	0.001		0.001	0.000	***	–0.003	0.002	
Psychographic	Risk taking	0.155	0.077	**	–0.094	0.047	**	0.036	0.049	
	Job satisfaction	–0.001	0.009		0.004	0.005		–0.009	0.006	
	Free time satisfaction	0.003	0.004		0.003	0.003		–0.004	0.003	
	Life satisfaction	–0.197	0.255		0.352	0.153	**	0.315	0.160	**
	Mood	0.001	0.002		–0.004	0.002	**	–0.002	0.002	
	Interest in politics	–0.323	0.115	***	0.364	0.067	***	0.152	0.076	**
Health	Care-dependent individuals in the household	–0.003	0.006		–0.001	0.003		–0.005	0.004	
	Chronic illness	–0.009	0.026		0.045	0.014	***	–0.009	0.016	
	Health insurance (compulsory)	0.340	0.113	***	–0.207	0.052	***	–0.121	0.059	**
Geographical	New states of Germany	0.125	0.016	***	–0.058	0.014	***	–0.017	0.013	
	Wald $\chi^2(147)$:				5531.99					
	Prob > χ^2				0.0000					
	Pseudo R^2				0.0532					

*Significance level: ***p<.01; **p<.05; *p<.10.*

will belong to this donor segment. Consistent with these results, respondents rely less on compulsory health insurance compared to non-donors, indicating that they may be family insurance or private insurance holders. Money and time donors have better (short-term) mood and are more interested in politics.

Blood and time donors. Blood and time donors reveal the least significant effects. The results show that these donors are younger, have no children, and are more interested in politics. In

Table 6. Elasticity Coefficients of Multinomial Logit Estimation (*Continued*)

Blood and Money Donation			*Money and Time Donation*			*Blood and Time Donation*			*Blood, Money, and Time Donation*		
Elasticity	*Standard error*	*Significance*	*Elasticity*	*Standard error*	*Significance*	*Elasticity*	*Standard error*	*Significance*	*Elasticity*	*Standard error*	*Significance*
–0.281	0.052	***	–0.059	0.030	*	0.067	0.053		–0.123	0.050	**
–0.918	0.239	***	0.916	0.139	***	–1.767	0.242	***	–0.068	0.231	
–0.045	0.024	*	0.018	0.014		0.015	0.018		0.037	0.021	*
0.070	0.059		–0.119	0.037	***	0.058	0.075		0.025	0.062	
–0.024	0.018		–0.053	0.013	***	–0.032	0.017	*	–0.048	0.021	**
0.108	0.039	***	0.178	0.027	***	–0.033	0.077		0.042	0.043	
0.061	0.070		0.117	0.040	***	–0.019	0.071		0.291	0.071	***
–0.050	0.035		0.129	0.020	***	–0.167	0.040	***	0.002	0.035	
0.009	0.005	*	0.005	0.003	*	0.001	0.007		0.002	0.006	
0.008	0.005		0.010	0.003	***	–0.005	0.007		0.016	0.004	***
–0.026	0.001	***	–0.001	0.001		0.001	0.001		0.000	0.001	
0.015	0.091		0.093	0.052	*	0.033	0.099		0.040	0.086	
0.007	0.010		0.001	0.006		–0.019	0.011	*	0.011	0.011	
0.007	0.006		–0.005	0.004		–0.010	0.006		–0.008	0.006	
0.031	0.330		0.239	0.196		–0.017	0.305		0.084	0.337	
0.003	0.002		–0.003	0.002	*	0.000	0.003		–0.001	0.002	
0.728	0.142	***	0.858	0.080	***	0.356	0.154	**	0.752	0.132	***
–0.006	0.008		–0.011	0.005	**	0.002	0.008		0.003	0.006	
0.068	0.030	**	0.019	0.017		–0.015	0.035		0.022	0.030	
–0.138	0.103		–0.277	0.058	***	0.064	0.128		–0.305	0.095	***
0.052	0.022	**	–0.062	0.016	***	0.087	0.023	***	–0.011	0.024	

addition, donors belonging to this segment are less satisfied with their jobs, indicating that volunteer work may be a way to compensate for job dissatisfaction.

Blood, money, and time donors. This group is made up of committed donors. Although our data do not contain information on donor motives, it seems logical to assume that this group may be driven by altruism and prosocial motivation. Typically, individuals from this segment are female and married. Receiving gifts from the social environment and being interested in politics increase the probability of belonging to this segment, which reflects their social engagement and prosocial attitude.

This study is the first to investigate donor segments based on their former engagement in different donation forms; the respective segment profiles therefore provide new findings and broaden the existing body of research.

Discussion and Implications

The results of our analysis are highly supportive of the possibility of identifying characteristics of donor segments involved in different donation forms related to time, blood, money, and their combinations. We apply a multinomial logistic model using non-donors as a reference category and identify donor profiles along sociodemographic, psychographic, health-related, and geographical aspects. Our segment categories are defined based on former donation behavior for money, blood, and time; however, we do not aim to develop a forecasting model that optimizes segment assignment based on our predictors. Instead, our objective is to investigate donor profiles and enrich the existing research on drivers of donation behavior and donor profiling.

We contribute to existing literature by simultaneously identifying profiles of donor segments basing on their former engagement in different forms of donation. To the best of our knowledge, this is the first study to examine all three donation forms simultaneously by defining segments of multiple donors and comparing these to donors of blood, money, or time as well as non-donors. We therefore expand the literature on donor profiling and the literature on donor relationship management, because our results deliver valuable insights for the customization of donor and (non-donor) activities related to new donor acquisition and retention. Our findings show that differences exist between subsegments of donors engaged in different donation forms. This implies that future studies on segment profiling should control for (former) behavior toward different donation forms to provide valid estimates for segment profiling. In addition, we identify significant effects from all four predictor categories used in our analyses, which underlines the need to capture for these effects in future studies.

Our analyses partially support the findings of previous research on drivers of the three donation formats and provide additional new insights. Consistent with prior research, we identify money donors as typically female, older, and with higher incomes (for example, Bekkers 2010; Clotfelter 2001; Priller and Schupp 2011; Rajan et al. 2009; Rosen and Sims 2011). Blood and time donors are typically not married and have no children, which is contrary to the earlier research findings (for example, Burnett 1981). Further, time donors are female and have lower incomes. Finally, we provide additional empirical results regarding variables with ambiguous effects in former studies, for example, age in case of blood donation (see Table 2).

Our findings have high managerial relevance. First, our segment definition based on former donation behavior corresponds to managerial practice of donor relationship managers or data miners. Still, existing research on multiple-donor profiling is missing, showing the need for more insights in this field. Because donors have limited amounts of time, money, and blood they may consider all three donation forms as substitutes. Given that donors who formerly engaged in one or more donation forms will evaluate the decision as to whether donate in an additional form differently from single-form donors or non-donors, NPOs should customize targeting strategies for these segments.

Specifically, for *blood donation services* we identify three relevant segments for donor retention: the *blood donor group*, the *blood and money donors,* and the *blood and time donors.* Specifically, *blood donors*—younger, unmarried people with no children and higher risk-taking profiles—should be addressed by mailings with bolder content. *Donors who (formerly) donated blood and money* are also younger than non-donors but differ from these and from single-blood donors regarding their financial background (higher incomes and higher probability of inheriting) and their frequency of chronic illnesses (see Table 7 and Appendix B). Targeting strategies motivating this group to re-donate blood can focus on communicating benefits of blood donation for the health status. We also discover a stronger interest in politics, leading to the implication that another targeting strategy for motivating (former) blood and money donors to reengage into blood donation should emphasize the high relevance of blood donation from a holistic perspective of health care system policies. Individuals who formerly engaged in *blood donation and volunteering* show lower job satisfaction. Hence, communication actions for this segment could focus on affective aspects of blood donation related to "warm glow."

For *NPOs engaged in monetary donations*, three segments are relevant for donor retention management: the *money donors*, *money and blood donors,* and *money and time donors.* We discover a higher amount of women within all three segments, leading to the implication that customer retention campaigns for these groups should target female segments, for example by appropriate mailing creations or layout. NPOs can customize their communication for *money donors*, for example, along the characteristic of satisfaction with life, which is generally higher within this segment. *Donors who (formerly) donated blood and money* are more frequently chronically ill. NPOs operating in the monetary donation sector can emphasize the flexibility of monetary donations, which are not related to health requirements and can be made even in times of illnesses. In addition, NPOs could use a health-related trigger in the communication, for example, by emphasizing that monetary donations can help people in need in times of illnesses. The *money and time donors* live in established familiar settings (older, married, have children) and have solid financial backgrounds (higher incomes, receiving gifts, and inheriting). Communication campaigns for this segment can be optimized, for example, by creating mailings based on family-related values and using corresponding images.

Voluntary organizations can use the characteristic of lower job satisfaction of *blood and time donors* and address this by drawing attention to the warm glow because of voluntary work.

Last, the *segment of committed donors* involved in all three donation formats is relevant for NPOs engaged in *blood donation, money donation,* as well as *voluntary work.* This segment is shown to consist of typically female individuals. Consequently, retention strategies should contain elements that address female potential donors as their primary target market, but new donor acquisition strategies could try to attract new male donors. We also discover a higher political interest within this segment. NPOs should emphasize the social relevance of donation and prosocial behavior for motivating individuals of this segment to re-donate.

Based on a large data set, our study has some limitations, of which we would like to point out three: First, our segment definition based on former donation behavior in the SOEP survey does not take into account the chronological order of the engagement in different donation forms. The structure of the SOEP data does not allow identifying the sequence of the engagements. Although the procedure of defining segments based on former donation behavior corresponds to best practice of donor relationship management and data mining, future research should control for the chronological order of engagement in different donation

Table 7. Overview of Segment Profiles

	Variable	*Blood Donation*	*Money Donation*	*Time Donation*	*Blood and Money Donation*	*Money and Time Donation*	*Blood and Time Donation*	*Blood, Money, and Time Donation*
Sociodemographic	Sex (male)		Female	Male	Female	Female		Female
	Age	Younger	Older	Younger	Younger	Older	Younger	
	Education/Apprenticeship			In education	Not in education			In education
	Full-time employment					Not full-time employed		
	Unemployment		Employed	Employed		Employed	Employed	Employed
	Net income		High net income	Low net income	High net income	High net income		
	Marital status (married)	Not married				Married		Married
	Children	No children	Children			Children	No children	
	Inheritance				Inheritance	Inheritance		
	Gift			Received no large gift		Received a large gift		Received a large gift
	Lottery winnings		Lottery winnings		No lottery winnings			
Psychographic	Risk-taking	Willing to take risks	Not willing to take risks			*Willing to take risks*		
	Job satisfaction						-	
	Free time satisfaction							
	Life satisfaction		Satisfied with life	Satisfied with life				
	Mood		Negative mood			Negative mood		
	Interest in politics	Not interested in politics	Interested in politics	Interested in politics	Interested in politics	Interested in politics	Interested in politics	Interested in politics
Health	Care-dependent individuals in the household					No care-dependent individuals in the household		
	Chronic illness		Chronically ill		Chronically ill			
	Health insurance (compulsory)	Compulsory insurance	No compulsory insurance	No compulsory insurance		No compulsory insurance		No compulsory insurance
Geographical	New states of Germany	New states of Germany	Old states of Germany		*New states of Germany*	Old states of Germany	*New States of Germany*	

Note: Specified characteristics increase the segment assignment probability compared to non-donors.

forms. Second, the donation motives are not available in the SOEP survey. Future studies should use enriched data comprising motivational information. Last, our sample is representative for Germany. We encourage further studies to extend our findings by investigating multiple donor profiles in other countries.

Notes

1. A detailed documentation of SOEP data is open to the public via the project's homepage (www.diw.de/soep).
2. See http://www.diw.de/en/diw_02.c.222729.en/questionnaires.html for the complete questionnaires of 2009 and 2010.
3. As short-time mood was shown to be a valid indicator even for the long term (Watson and Walker 1996), we expect this variable to account for variation among donors and non-donors, for example, because of the warm glow effect (Andreoni 1990).

References

Andaleeb, S. S., and A. K. Basu. 1995. "Explaining Blood Donation: The Trust Factor." *Journal of Health Care Marketing* 15 (1): 42–48.

Andreoni, J. 1990. "Impure Altrusim and Donations to Public-Goods—A Theory of Warm-Glow Giving." *Economic Journal* 100 (401): 464–77.

Banks, J., and S. Tanner. 1999. "Patterns in Household Giving: Evidence from U.K. Data." *Voluntas* 10 (2): 167–78.

Bekkers, R. 2006. "Traditional and Health-Related Philanthropy: The Role of Resources and Personality." *Social Psychology Quarterly* 69 (4): 349–66.

———. 2010. "Who Gives What and When? A Scenario Study of Intentions to Give Time and Money." *Social Science Research* 39 (3): 369–81.

Berger, I. E. 2006. "The Influence of Religion on Philanthropy in Canada." *Voluntas* 17 (2): 110–27.

Boulware, L. E., L. E. Ratner, P. M. Ness, L. A. Cooper, S. Campbell-Lee, and T. A. LaVeist. 2002. "The Contribution of Sociodemographic, Medical, and Attitudinal Factors to Blood Donation among the General Public." *Transfusion* 42 (6): 669–78.

Bryant, W. K. 2003. "Participation in Philanthropic Activities: Donating Money and Time." *Journal of Consumer Policy* 26 (1): 43–73.

Burnett, J. J. 1981. "Psychographic and Demographic Characteristics of Blood Donors." *Journal of Consumer Research* 8 (1): 62–66.

Burnett, J. J., and J. H. Leigh. 1986. "Distinguishing Characteristics of Blood Donor Segments Defined in Terms of Donation Frequency." *Journal of Health Care Marketing* 6 (2): 38–48.

Clotfelter, C. T. 2001. "Who Are the Alumni Donors? Giving by Two Generations of Alumni from Selective Colleges." *Nonprofit Management and Leadership* 12 (2): 119–38.

Duboz, P., and B. Cunéo. 2009. "Impact of Socioeconomic Status on Blood Donation." *Transfusion Clinique Et Biologique: Journal De La Société Française De Transfusion Sanguine* 16 (4): 371–78.

Duboz, P., and B. Cunéo. 2010. "How Barriers to Blood Donation Differ between Lapsed Donors and Non-donors in France." *Transfusion* 20 (4): 227–36.

Duboz, P., E. Macia, and B. Cunéo. 2010. "Sociodemographic and Attitudinal Factors to Blood Donation in the Urban Population of Dakar, Senegal." *Transfusion* 50 (12): 2713–20.

Eikenberry, A. M. 2009. "Refusing the Market: A Democratic Discourse for Voluntary and Nonprofit Organizations." *Nonprofit and Voluntary Sector Quarterly* 38 (4): 582–96.

Einolf, C. J. 2011. "Gender Differences in the Correlates of Volunteering and Charitable Giving." *Nonprofit and Voluntary Sector Quarterly* 40 (6): 1092–1112.

Feld, S., H. Frenzen, M. Krafft, K. Peters, and P. C. Verhoef. 2013. "The Effects of Mailing Design Characteristics on Direct Mail Campaign Performance." *International Journal of Research in Marketing* 30 (2): 143–59.
Greene, W. H. 2003. *Econometric Analysis.* Vol. 5. Upper Saddle River, NJ: Prentice Hall.
Hermes, V. 2007. "Das ist ein super Job, den ich hier habe!" [This is a Great Job I Have Here!] *Absatzwirtschaft* 50 (11): 100–03.
Houston, D. J. 2006. "'Walking the Walk' of Public Service Motivation: Public Employees and Charitable Gifts of Time, Blood, and Money." *Journal of Public Administration Research and Theory* 16 (1): 67.
Hughes, P., and W. Luksetich. 2008. "Income Volatility and Wealth: The Effect on Charitable Giving." *Nonprofit and Voluntary Sector Quarterly* 37 (2): 264–80.
Ibrahim, N. A., and M. F. Mobley. 1993. "Recruitment and Retention of Blood Donors: A Strategic Linkage Approach." *Health Care Management Review* 18 (3): 67–73.
Jeong, H. O. 2010. "How Do Religions Differ in Their Impact on Individuals' Social Capital? The Case of South Korea." *Nonprofit and Voluntary Sector Quarterly* 39 (1): 142–60.
Jones, K. S. 2006. "Giving and Volunteering as Distinct Forms of Civic Engagement: The Role of Community Integration and Personal Resources in Formal Helping." *Nonprofit and Voluntary Sector Quarterly* 35 (2): 249–66.
Kottasz, R. 2004. "Differences in the Donor Behavior Characteristics of Young Affluent Males and Females: Empirical Evidence from Britain." *Voluntas* 15 (2): 181–203.
Krupp, H.-J. 2007. *Das Sozio-oekonomische Panel (SOEP): Genese und Implementation.* [The Socio-Economic Panel (SOEP): Genesis and Implementation] SOEP papers on Multidisciplinary Panel Data Research. DIW. Berlin.
Lee, L., J. A. Piliavin, and V. R. A. Call. 1999. "Giving Time, Money, and Blood: Similarities and Differences." *Social Psychology Quarterly* 62 (3): 276–90.
Mesch, D. J., P. M. Rooney, K. S. Steinberg, and B. Denton. 2006. "The Effects of Race, Gender, and Marital Status on Giving and Volunteering in Indiana." *Nonprofit and Voluntary Sector Quarterly* 35 (4): 565–87.
Meslin, E. M., P. M. Rooney, and J. G. Wolf. 2008. "Health-Related Philanthropy: Toward Understanding the Relationship between the Donation of the Body (and Its Parts) and Traditional Forms of Philanthropic Giving." *Nonprofit and Voluntary Sector Quarterly* 37 (1 suppl): 44S–62S.
Monin, B., and D. T. Miller. 2001. "Moral Credentials and the Expression of Prejudice." *Journal of Personality and Social Psychology* 81 (1): 33–43.
Mount, J. 1996. "Why Donors Give." *Nonprofit Management and Leadership* 7 (1): 3–14.
Naskrent, J., and P. Siebelt. 2011. "The Influence of Commitment, Trust, Satisfaction, and Involvement on Donor Retention." *Voluntas* 22 (4): 757–78.
Newman, K., and T. Pyne. 1997. "Service Quality and Blood Donors—A Marketing Perspective." *Journal of Marketing Management* 13 (6): 579–99.
Nguyen, D. D., D. A. Devita, N. V. Hirschler, and E. L. Murphy. 2008. "Blood Donor Satisfaction and Intention of Future Donation." *Transfusion* 48 (4): 742–48.
Nonis, S. A., et al. 1996. "College Student's Blood Donation Behavior: Relationship to Demographics, Perceived Risk, and Incentives." *Health Marketing Quarterly* 13 (4): 33–46.
Oswalt, R. M., and M. Napoliello. 1974. "Motivations of Blood Donors and Nondonors." *Journal of Applied Psychology* 59 (1): 122–24.
Ownby, H. E., F. Kong, K. Watanabe, Y. Tu, and C. C. Nass. 1999. "Analysis of Donor Return Behavior: Retrovirus Epidemiology Donor Study." *Transfusion* 39 (10): 1128–35.
Piper, G., and S. Schnepf. 2008. "Gender Differences in Charitable Giving in Great Britain." *Voluntas* 19 (2): 103–24.
Priller, E., and J. Schupp. 2011. "Soziale und ökonomische Merkmale von Geld- und Blutspendern in Deutschland." [Social and Economic Characteristics of Money and Blood Donors in Germany] *DIW Wochenbericht* 78 (29): 3–10.
Rajan, S. S., G. H. Pink, and W. H. Dow. 2009. "Sociodemographic and Personality Characteristics of Canadian Donors Contributing to International Charity." *Nonprofit and Voluntary Sector Quarterly* 38 (3): 413–40.

Reed, P. B. 2000. "Distinguishing Characteristics of Active Volunteers in Canada." *Nonprofit and Voluntary Sector Quarterly* 29 (4): 571–92.

Rosen, H. S., and S. T. Sims. 2011. "Altruistic Behavior and Habit Formation." *Nonprofit Management and Leadership* 21 (3): 235–53.

Sargeant, A. 2001. "Relationship Fundraising: How to Keep Donors Loyal." *Nonprofit Management and Leadership* 12 (2): 177–92.

Sargeant, A., J. Hudson, and D. C. West. 2008. "Conceptualizing Brand Values in the Charity Sector: The Relationship between Sector, Cause and Organization." *Service Industries Journal* 28 (5): 615–32.

Sargeant, A., and L. Woodliffe. 2005. "The Antecedents of Donor Commitment to Voluntary Organizations." *Nonprofit Management and Leadership* 16 (1): 61–78.

Schervish, P. G., and J. J. Havens. 1997. "Social Participation and Charitable Giving: A Multivariate Analysis." *Voluntas* 8 (3): 235–60.

Schreiber, G. B., S. A. Glynn, M. A. Damesyn, D. J. Wright, Y. Tu, and R. Y. Dodd. 2003. "Lapsed Donors: An Untapped Resource." *Transfusion* 43 (1): 17–24.

Schreiber, G. B., U. K. Sharma, D. J. Wright, S. A. Glynn, H. E. Ownby, and Y. Tu. 2005. "First Year Donation Patterns Predict Long-Term Commitment for First-Time Donors." *Vox Sanguinis* 88 (2): 114–21.

Schupp, J., and G. Wagner. 2002. "Maintenance of and Innovation in Long-Term Panel Studies: The Case of the German Socio-Economic Panel (GSOEP)." DIW Discussion Paper 276: 1–16.

Solomon, G. 2012. "Segmentation and Communications to Solve the Blood Shortage: An Exploration of the Problem with Recommendations." *Voluntas* 23 (2): 415–33.

Sosin, M. R. 2012. "Social Expectations, Constraints, and Their Effect on Nonprofit Strategies." *Nonprofit and Voluntary Sector Quarterly* 41 (6): 1231–50.

Taniguchi, H. 2010. "Who Are Volunteers in Japan?" *Nonprofit and Voluntary Sector Quarterly* 39 (1): 161–79.

Taniguchi, H., and L. Thomas. 2011. "The Influences of Religious Attitudes on Volunteering." *Voluntas* 22 (2): 335–355.

Thomson, R. A., J. Bethel, A. Y. Lo, H. E. Ownby, C. C. Nass, and A. E. Williams. 1998. "Retention of 'Safe' Blood Donors." *Transfusion* 38 (4): 359–67.

Tscheulin, D. K., and J. Lindenmeier. 2005. "The Willingness to Donate Blood: An Empirical Analysis of Sociodemographic and Motivation-Related Determinants." *Health Services Management Research* 18 (3): 165–74.

Veldhuizen, I. J. T., C. J. M. Doggen, F. Atsma, and W. L. A. M. De Kort. 2009. "Donor Profiles: Demographic Factors and Their Influence on the Donor Career." *Vox Sanguinis* 97 (2): 129–38.

Venable, B. T., G. M. Rose, V. D. Bush, and F. W. Gilbert. 2005. "The Role of Brand Personality in Charitable Giving: An Assessment and Validation." *Journal of the Academy of Marketing Science* 33 (3): 295–312.

Watson, D., and L. M. Walker. 1996. "The Long-Term Stability and Predictive Validity of Trait Measures of Affect." *Journal of Personality and Social Psychology* 70 (3): 567–77.

Wiepking, P. 2007. "The Philanthropic Poor: In Search of Explanations for the Relative Generosity of Lower Income Households." *Voluntas* 18 (4): 339–58.

Wiepking, P., and R. Bekkers. 2010. "Does Who Decides Really Matter? Causes and Consequences of Personal Financial Management in the Case of Larger and Structural Charitable Donations." *Voluntas* 21 (2): 240–63.

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Appendix A. Descriptive Statistics

	Variables	*Frequency*	*Sample*	*Mean*	*SE*
Donation	Blood donation	19.66%	*N* = 11,033		
	Money donation	38.94%	*N* = 11,000		
	Time donation	32.06%	*N* = 12,412		
Sociodemographic	Sex (male)	49.40%	*N* = 12,487		
	Age		*N* = 12,487	46.787	0.125
	Education/Apprenticeship	11.04%	*N* = 12,487		
	Full-time employment	47.91%	*N* = 12,486		
	Unemployment	7.08%	*N* = 12,487		
	Net income		*N* = 12,437	1216.242	12.549
	Marital status (married)	61.05%	*N* = 12,487		
	Children	31.25%	*N* = 12,487		
	Inheritance	1.83%	*N* = 12,487		
	Gift	2.31%	*N* = 12,487		
	Lottery winnings	0.20%	*N* = 12,487		
Psychographic	Risk taking		*N* = 12,456	4.507	0.020
	Job satisfaction		*N* = 12,487	0.133	0.008
	Free time satisfaction		*N* = 12,487	−0.106	0.008
	Life satisfaction		*N* = 12,463	7.175	0.015
	Mood		*N* = 12,444	−0.044	0.009
	Interest in politics		*N* = 12,439	2.316	0.007
Health	Care-dependent individuals in the household	2.29%	*N* = 12,487		
	Chronic illness	28.93%	*N* = 12,455		
	Health insurance (compulsory)	84.14%	*N* = 12,475		
Geographical	New states of Germany	21.41%	*N* = 12,487		

Appendix B. Descriptive Statistics for Donor Profiles

		Non-donors		Blood Donation		Money Donation		Time Donation		Blood and Money Donation		Money and Time Donation		Blood and Time Donation		Blood, Money, and Time Donation	
	Variable	*Mean*	*SD*	*Mean*	*SD*	*Mean*	*SD*	*Mean*	*SD*	*Mean*	*SD*	*Mean*	*SD*	*Mean*	*SD*	*Mean*	*SD*
Sociodemographic	Sex (male)	49.25%		49.04%		**44.94%**		**53.39%**		**42.19%**		**51.07%**		55.56%		**50.31%**	
	Age	45.78	14.20	**39.61**	13.25	**51.11**	12.67	**45.25**	14.44	**45.67**	12.84	**50.39**	12.57	**40.29**	13.88	47.94	12.66
	Education/ Apprenticeship	11.26%		21.92%		5.32%		**15.48%**		**8.72%**		7.16%		23.19%		**12.19%**	
	Full-time employment	44.99%		47.95%		50.69%		47.46%		56.80%		**51.07%**		51.20%		52.81%	
	Unemployment	10.54%		9.59%		**3.89%**		**4.84%**		3.85%		**2.51%**		**5.07%**		2.50%	
	Net income	1,005.48	1,162.05	1,040.96	937.07	1,527.18	1,870.32	**1,155.91**	1,251.02	**1,469.26**	1,291.33	**1,609.27**	1,706.04	1,120.89	1,067.87	1,467.25	1,248.95
	Marital status (married)	56.23%		**42.74%**		70.18%		60.22%		62.27%		**74.25%**		45.17%		**72.50%**	
	Children	30.06%		**29.59%**		**30.14%**		33.37%		32.05%		**36.49%**		**24.15%**		31.86%	
	Inheritance	1.51%		1.10%		2.13%		1.69%		**2.84%**		**2.45%**		1.93%		2.81%	
	Gift	1.54%		3.01%		2.45%		1.87%		3.85%		**4.27%**		2.66%		**4.37%**	
	Lottery winnings	0.09%		0.55%		0.53%		0.06%		**0.00%**		0.25%		0.48%		0.31%	
Psychographic	Risk taking	4.46	2.31	**4.93**	2.29	4.31	2.23	4.62	2.28	4.54	2.13	**4.60**	2.11	4.79	2.15	4.55	2.02
	Job satisfaction	.04	.98	.15	.93	.23	.83	.15	.95	.28	.89	.26	.84	.11	.92	.32	.79
	Free time satisfaction	–.12	.95	–.21	.97	–.12	.93	–.08	.92	–.19	.99	–.04	.88	–.08	.86	–.06	.85
	Life satisfaction	6.99	1.78	7.16	1.70	7.33	1.53	**7.27**	1.64	7.35	1.60	7.44	1.56	7.17	1.74	7.47	1.56
	Mood	–.03	1.01	–.08	.98	–.01	.95	–.04	.96	–.10	.96	**–.09**	.91	–.05	.97	–.10	.98
	Interest in politics	2.14	.78	**2.15**	.714	2.47	.74	**2.37**	.79	**2.47**	.74	**2.62**	.76	**2.37**	.77	**2.57**	.74
Health	Care–dependent individuals in the household	2.76%		1.64%		2.34%		1.75%		1.62%		**1.57%**		1.93%		2.19%	
	Chronic illness	28.03%		17.86%		33.78%		27.41%		**30.49%**		32.41%		24.27%		23.44%	
	Health insurance (compulsory)	89.78%		**89.84%**		77.00%		**83.59%**		80.32%		73.33%		86.96%		**76.56%**	
Geographical	New states of Germany	23.39%		37.81%		**16.08%**		20.07%		**24.54%**		15.95%		**29.47%**		16.88%	

Note: Bold = significant effect for respective profile.